



MERLIN IP

Technical Specification

1. System Overview

MERLIN IP for board-to-board time synchronization over long distances. The synchronization system enables multiple embedded boards (FPGA/SoC/DSP platforms) to operate with deterministic timing alignment using hardware synchronization methods over optical fiber.

2. Reference Architecture

Master Time Source:

- GNSS disciplined clock (GPS/GLONASS/Galileo/BeiDou)
- IRIG-B or 1PPS disciplined oscillator

Distribution Methods:

- 1PPS signal distribution
- 10 MHz frequency reference distribution
- IRIG time code distribution

3. Hardware Synchronization Specifications

- Clock Resolution: ≤ 5 ns internal resolution.
- 1PPS Accuracy: ± 15 ns to ± 50 ns to GNSS reference.
- Frequency Reference: 10 MHz sine/LVCMOS disciplined output.
- Oscillator Options: TCXO / OCXO (stability up to 1×10^{-12}).
- Holdover Performance: Maintains phase continuity during reference loss.
- Event Time Tagging: Up to 10,000 events/sec with nanosecond resolution.
- PLL lock Time: Transceiver PLL lock to reference is very short ($\sim \leq 1$ ms typical).
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4. Board-Level Integration Requirements

- Differential or single-ended 1PPS input with impedance matching
- 10 MHz low-jitter clock input to PLL/MMCM (FPGA)
- FPGA logic for time counter reset on PPS rising edge
- Hardware timestamp capture registers for event tagging
- Optional PTP stack with hardware MAC timestamping

5. Synchronization Performance Targets

- Inter-board phase alignment: ≤ 20 ns (hardware sync)
- Deterministic trigger skew: ≤ 10 ns
- Network sync accuracy (PTP): ≤ 1 μ s typical



- Frequency stability: $\leq 1 \times 10^{-12}$

6. Operational Modes

- GNSS Disciplined Mode: Can accept an external clock
- IRIG-B Disciplined Mode: Can accept an external clock
- Holdover Mode
- Free-running Mode

7. Typical Synchronization Workflow

1. GNSS reference disciplines master clock.
2. The master distributes 1 PPS and 10 MHz signals to all boards.
3. Each board locks the internal PLL to a 10 MHz reference.
4. 1PPS resets or aligns time counters simultaneously.
5. Optional PTP maintains network-level synchronization.



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